Introduction to Database Management Systems (DBMS)

What is DBMS?

A Database Management System (DBMS) is software used to create, store, manage, and retrieve data efficiently.

Examples:

- MySQL
- PostgreSQL
- Oracle
- SQL Server

Advantages of DBMS

1. Reduces data redundancy
2. Improves data consistency
3. Provides security
4. Supports backup and recovery
5. Enables concurrent access

Database Models

Relational Model

Data is stored in tables consisting of rows and columns.

Example:

Student Table

ID	Name	Department
1	John	CSE
2	Alice	ECE

Primary Key

A Primary Key uniquely identifies each record in a table.

Example:

Student_ID

Foreign Key

A Foreign Key creates a relationship between two tables.

SQL

SQL (Structured Query Language) is used to communicate with databases.

Common Commands:

SELECT – Retrieve data

INSERT – Add data

UPDATE – Modify data

DELETE – Remove data

Normalization

Normalization is the process of organizing data to reduce redundancy and improve integrity.

Normal Forms:

1NF – First Normal Form

2NF – Second Normal Form

3NF – Third Normal Form

ACID Properties

Atomicity – Transactions are completed fully or not at all.

Consistency – Database remains valid.

Isolation – Transactions do not interfere.

Durability – Changes are permanently stored.

Conclusion

DBMS is essential for efficient data management and is widely used in modern applications.